

Dataset Inspection Report

Telco Customer Churn Dataset — Data Analytics Internship, AnalystLab Africa

1. Dataset Overview

The dataset used for this project is the Telco Customer Churn dataset (file: WA_Fn-UseC_-Telco-Customer-Churn.csv). It contains one record per customer for a telecommunications company, capturing demographic details, account information, subscribed services, and whether the customer churned. The dataset was imported into Excel using Data > Get Data > From Text/CSV to ensure the comma-delimited columns were read correctly.

Property	Value
Number of rows (customers)	7,043
Number of columns	21
Unit of observation	One row per customer
Target variable	Churn (Yes / No)

2. Column Data Types

Each of the 21 columns was checked by inspecting cell alignment (numbers align right; text aligns left by default in Excel). Most columns are categorical (text) fields such as gender, Contract, InternetService, and PaymentMethod. Three columns are numeric: tenure (months), MonthlyCharges, and TotalCharges.

One inconsistency was identified: TotalCharges, although representing a monetary amount, was stored as a text field rather than a number, confirmed by its left-aligned values. This required special handling during inspection (see Section 3).

3. Missing Values

A standard blank check using COUNTBLANK() was run across all 21 columns and returned 0 for every column, including TotalCharges — suggesting the dataset was complete.

However, because TotalCharges was stored as text rather than numeric, a deeper check was carried out using a TRIM-based formula (SUMPRODUCT(--(TRIM(range)=""))), which detects blank-looking text values that COUNTBLANK does not catch. This revealed 11 hidden blank entries in TotalCharges that a standard check missed entirely.

Likely Explanation

These 11 rows are believed to belong to customers with zero months of tenure — i.e. customers who had only just joined and had not yet received a bill. This is a plausible, logical explanation for the gap rather than a data entry error, though it was not exhaustively cross-verified row by row in this analysis.

Recommended Treatment

1. Convert both MonthlyCharges and TotalCharges from text to numeric data types (e.g. using the VALUE() function), since neither can be used in calculations or charts while stored as text.
2. Fill the 11 hidden blank entries in TotalCharges with 0, consistent with new customers who have no billing history yet — MonthlyCharges requires no such fix, as it has no missing values.

4. Duplicate Records

Duplicate rows were checked using Excel's Data > Remove Duplicates tool across all columns, and customerID uniqueness was independently verified using conditional formatting to visually flag any repeated IDs.

Both checks confirmed there are no duplicate records in the dataset: all 7,043 customerID values are unique, and no full-row duplicates were found.

5. Summary Statistics (Numeric Fields)

Descriptive statistics were computed for the three key numeric fields using AVERAGE, MEDIAN, MIN, MAX, and STDEV.S.

Field	Average	Median	Min	Max	Std Dev
tenure (months)	32.37	29.00	0	72	24.56
MonthlyCharges (\$)	65.30	71	18.25	116	30.09
TotalCharges (\$)	2323.	1,394.55	19	8405	2224.94

Note: statistics for TotalCharges were calculated after first converting the column to a numeric data type, since formulas such as AVERAGE cannot compute correctly on text-formatted values.

Key observations: tenure ranges from 0 to 72 months with an average of roughly 32 months, indicating a wide mix of new and long-standing customers. Monthly charges range from \$18.25

to \$118.75, and the gap between the average and median for TotalCharges suggests the distribution is right-skewed, with a smaller group of long-tenured, high-spend customers pulling the average upward.

Note: the minimum value shown for TotalCharges (\$19.00) reflects only the 7,032 rows with valid numeric entries; it excludes the 11 rows with hidden blank values discussed in Section 3.

6. Categorical Field Cardinality

Category counts were checked using PivotTables built on each categorical column. Most fields (e.g. Partner, Dependents, PhoneService, PaperlessBilling) are binary (Yes/No), while service-related fields such as InternetService, Contract, and PaymentMethod have 3-4 categories. For example, Contract was confirmed to contain exactly three categories: Month-to-month, One year, and Two year.

This low cardinality makes the dataset well-suited to bar chart and pie chart visualisation without excessive fragmentation.

7. Conclusion

After inspection, the dataset comprises 7,043 rows and 21 columns with no duplicate records and no missing values beyond the 11 hidden blanks identified in TotalCharges. Once TotalCharges is converted to a numeric type and its blanks addressed, the dataset is fit for exploratory analysis and reporting on customer churn drivers.